

Topic 1.13 - Function Model Selection and Assumption Articulation

Guided Notes

Strengthened ECHS edition - reasoning, challenge, and AP-style practice

Name: _____

Class: _____

Date: _____

Why this topic matters

Many models can fit a small data set. A defensible choice combines patterns in differences or ratios, residual behavior, contextual mechanisms, and a realistic domain. Equally important, assumptions and limitations must be stated so that a mathematical fit is not mistaken for certainty outside the conditions under which the data were collected.

Learning targets

- Use differences, ratios, rates, and graphical shape to compare candidate model families.
- Compute and interpret residuals and use residual patterns to evaluate fit.
- State assumptions, contextual domains, interpolation limits, and extrapolation risks in precise language.

Language and structure

Term	Meaning in this topic
residual	Observed value minus model-predicted value.
interpolation	Prediction within the span of observed inputs.
extrapolation	Prediction beyond the observed input range.
assumption	A condition treated as true so the model can be used.
model adequacy	Whether a model is accurate and appropriate enough for its intended decision.

Launch - reason before calculating

Two models have nearly identical coefficients of determination for a data set. Explain why that statistic alone cannot determine which model should be used for a ten-year prediction.

Concept 1: Structural evidence from differences and ratios

Core idea

Approximately constant first differences support a linear model; approximately constant second differences at equally spaced inputs support a quadratic model; approximately constant ratios support an exponential model. These patterns are evidence, not automatic proof, especially with noisy or sparse data.

Worked example - annotate the reasoning

Problem. For $x = 0, 1, 2, 3, 4$, outputs are 3, 7, 15, 27, 43. Select a model family and construct an exact model.

Model reasoning. First differences are 4, 8, 12, 16 and second differences are constant at 4, so use a quadratic with $2a = 4$, giving $a = 2$. Solving gives $y = 2x^2 + 2x + 3$.

Check your understanding

Independent

No calculator

Outputs at equally spaced inputs are 5, 7.5, 11.25, 16.875. Select a model family and write a model beginning at input 0.

Concept 2: Residuals as a diagnostic, not decoration

Core idea

A residual is $y - \hat{y}$. Positive residuals mean the model underpredicted; negative residuals mean it overpredicted. A suitable model tends to produce small residuals scattered without a systematic curve, trend, or change in spread. A high R^2 can coexist with a problematic residual pattern.

Worked example - annotate the reasoning

Problem. Model A has residuals $-4, 1, 5, 2, -4$; Model B has $0.2, -0.3, 0.1, 0.2, -0.2$. Compare the models.

Model reasoning. Model B has much smaller residuals and no strong visible pattern. Model A shows large edge errors and a middle bulge, suggesting missing curvature. Model B is better on this evidence.

Check your understanding

Independent

No calculator

An observed value is 42.7 and the model predicts 45.1. Find and interpret the residual.

Concept 3: Context, assumptions, and responsible prediction

Core idea

A model is used only on a defensible domain and under stated assumptions. Identify what is held stable, how the data were measured, whether inputs are continuous or discrete, and whether outside forces may change. Interpolation is usually less risky than extrapolation, but neither is guaranteed.

Worked example - annotate the reasoning

Problem. A medicine-decay model was fitted for the first 12 hours in healthy adults. A student predicts concentration after 10 days. Evaluate the decision.

Model reasoning. The prediction is extreme extrapolation and ignores biological changes, measurement limits, dosing, and population differences. The fitted equation may be mathematically evaluable but not scientifically defensible there.

Check your understanding

Independent

No calculator

A quadratic revenue model is based on prices from 20 to 50 riyals. State two assumptions needed to use it at 45 riyals and one reason not to use it at 500 riyals.

Representation laboratory

Connect at least two representations

For inputs 0, 1, 2, 3, 4, a data set has outputs 6, 9, 13.5, 20.25, 30.375. Represent the pattern using a table of ratios, an equation, and a verbal interpretation of the parameter 1.5.

Challenge laboratory

Challenge - justify every claim

On $0 \leq x \leq 4$, the models $L(x) = 3x + 2$ and $Q(x) = 0.05x^2 + 2.8x + 2.1$ produce similar values. Compare their predictions at $x = 40$ and explain what the comparison teaches about extrapolation.

Error clinic and calculator strategy

Common errors to diagnose

- Choosing the model with the largest R^2 while ignoring a patterned residual plot.
- Calling a relationship causal because a regression fits well.
- Using a continuous domain for a count of objects without explanation.
- Reporting an extrapolated value without discussing whether assumptions remain plausible.

Technology decision

Technology can compute regressions and residual plots quickly. Enter data carefully, inspect axes and residual scale, and compare models using both numerical and contextual evidence. Save enough precision for calculations, then round only the final reported result appropriately.

AP-style synthesis**Multi-part reasoning task**

A data set has inputs 0, 1, 2, 3, 4, 5 and observed outputs 4.1, 7.0, 14.2, 24.8, 40.1, 59.0. Quadratic predictions are 4, 7, 14, 25, 40, 59; exponential predictions are 4.8, 7.5, 11.7, 18.3, 28.6, 44.7.

A. Compute the residual at input 3 for each model and interpret its sign.

B. Select the more defensible model using the full residual evidence.

C. State one assumption and one limitation if the selected model is used to predict at input 8.

Exit ticket**Before you leave**

1. What does a negative residual mean?

2. Which difference pattern suggests a quadratic model?

3. Why is extrapolation usually riskier than interpolation?
